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#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>
#include <assert.h>

#define NELEM 80000000

long v[NELEM];
int prim = 0;

void *busca_inc(void *arg) {
    long i, ret = -1, n = (long)arg;

    for (i = 0; i < NELEM; i++) {
        if (v[i] == n) {
            ret = i;
            break;
        }
    }
    if (prim == 0)
        prim = 1;
    pthread_exit((void *)ret);
}

void *busca_dec(void *arg) {
    long i, ret = -1, n = (long)arg;

    for (i = NELEM-1; i >= 0; i--) {
        if (v[i] == n) {
            ret = i;
            break;
        }
    }
    if (prim == 0)
        prim = 2;
    pthread_exit((void *)ret);
}

int main(int argc, char *argv[])
{
    long i, n;
    pthread_t thr, thr2;
    int rc;
    void *status;

    if (argc < 2) {
        fprintf(stderr, "e o n, querido?\n");
        exit(1);
    }
    n = atoi(argv[1]);
    for (i = 0; i < NELEM; i++)
        v[i] = i+1;
    //i = busca_inc(n);
    rc = pthread_create(&thr, NULL, busca_inc, (void *)n); assert(rc == 0);
    rc = pthread_create(&thr2, NULL, busca_dec, (void *)n); assert(rc == 0);
    rc = pthread_join(thr, &status); assert(rc == 0);
    rc = pthread_join(thr2, &status); assert(rc == 0);
    i = (long)status;
    if (i >= 0)
        printf("O numero %ld esta' na posicao %ld do vetor (prim=%d)\n",
            n, i, prim);
    else
        printf("O numero %ld nao foi encontrado no vetor (prim=%d)\n", n,
            prim);
    return 0;
}
```